

Project proposal - Solving airport schedules

Emil Ulvagården

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1 Objective

1.1 Which domain and the problem you will address?

There are two domains:

- The air traffic control domain for aircraft arrivals, taxiing, gates and departures
- The scheduling optimization domain for allocating limited gates and taxiways efficiently while respecting time constraints

The problem getting addressed is that airports have a finite number of gates and a limited taxiway capacity, while each aircraft have a arrival and departure time window. The challenge is to assign each aircraft to an appropriate gate and time slot such that:

- All aircraft are accommodated within their arrival/departure time intervals
- No gate/taxiway is over capacity

2 Models

2.1 What models and metamodels are needed?

I foresee the use of five metamodels:

- One metamodel that represents the Flight information and time windows (FlightMM)
- One metamodel that represents the Airport infrastructure (AirportMM)
- A metamodel that represents the Scheduling (ScheduleMM)
- Two metamodels to interact with third party tools used for solving the schedule:
 - One metamodel representing the solver input (SolverInputMM)
 - One metamodel representing the solver output (SolverResultMM)

2.2 How many domains, MMs, and their relations.

There are four domains:

- Flight
- Airport infrastructure
- Scheduling
- Third-party schedule solver

The FlightMM and AirportMM are independent but provide information for gate allocation.

The ScheduleMM relates to both FlightMM and AirportMM by defining the assignment so linking flights to gates and the scheduling constraints.

The SolverInputMM is generated through a M2M transformation using FlightMM, AirportMM and ScheduleMM and represents the model passed to the external solver.

The SolverResultMM represents the solver's output and is used to update the schedule model through another M2M transformation

2.3 Tool integration - external tools that produce or consume models. model compliance

Integration of third-party tools such as Google OR-Tools can be used to compute the optimal gate assignment based on the constraints of gate availability and the time windows for arrival and departure. The SolverResult model is created from the solver output using a T2M transformation. The Schedule model is then updated through an M2M transformation.

2.4 How will the model be created, updated?

The Flight model could be created either manually or automatically with existing data in csv format from ex [1] with adding interval for arrival/departure time. The data would then need to be updated manually unless other sources can be found.

The Airport model will be created manually and updated manually if the information is to change.

The Schedule model will be automatically generated through M2M transformation from Flight and Airport models.

The SolverInput model will be automatically created from Schedule, Flight and Airport models.

The SolverResults will be created through a T2M transformation from the solver output and an M2M transformation will then update the Schedule model with the optimized schedule.

3 Transformations

3.1 What transformation will you develop?

Six transformations will be developed:

1. M2M using Flight and Airport to generate the Schedule, this will generate an initial schedule linking flights to gates within their time window
2. M2M using the Schedule to generate the SolverInput for the optimization tool
3. M2T using the SolverInput to produce an input file for the solver
4. T2M output file from the solver to the SolverResult
5. M2M from SolverResult to update Schedule to reflect the optimal schedule assignment
6. M2T from Schedule to a text file for a readable schedule.

3.2 How will you combine transformations?

The M2M using Flight and airport to generate the Schedule execute in sequence with the M2M Schedule to SolverInput and the M2T SolverInput to input file. This is to produce the necessary inputs to the external schedule solver.

The T2M solver output to SolverResult and the M2M SolverResult to Schedule execute sequentially since their combined purpose is to update the schedule.

The M2T from Schedule to text file execute when the schedule has been updated independently to create a view of the updated schedule for all flights.

4 Figure

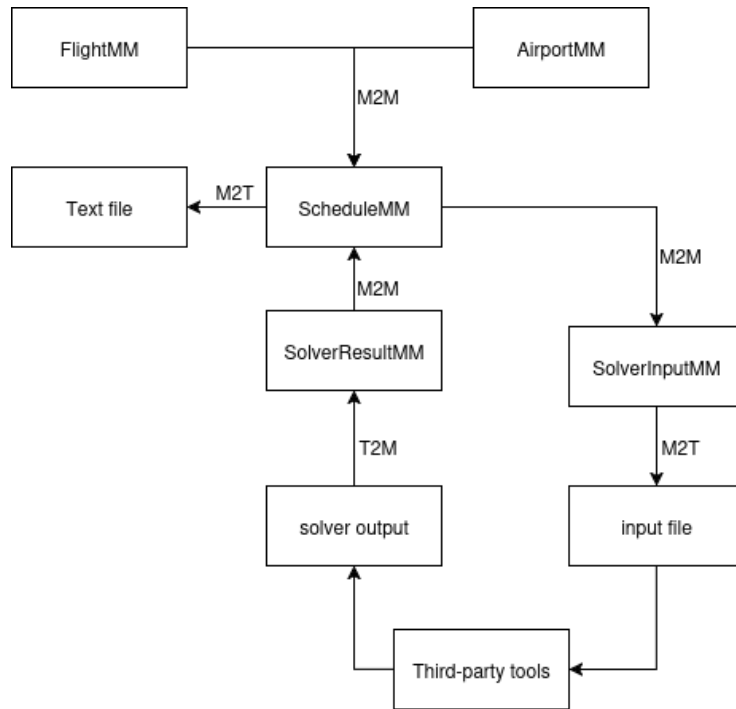


Figure 1: Model creation and update

5 References

1. <https://github.com/davidmegginson/ourairports-data>
2. <https://developers.google.com/optimization>